Java Programming – Classes, Objects, Constructors and Methods (5 Programs)

1.Create a class Student with student ID, name and age. Create three student objects and display their details.

Code:

```
class Student {  int st_Id;     String name;     int age;
   void displayDetails() {
      System.out.println("Student ID: " + st_Id);
      System.out.println("Name: " + name);
      System.out.println("Age: " + age);
      System.out.println("--------------------"); }
public static void main(String[] args) {
      Student s1 = new Student();
      s1.st_Id = 101; s1.name = "Rahul"; s1.age = 20;
      Student s2 = new Student();
      s2.st_Id = 102;  s2.name = "Priya";  s2.age = 19;
      Student s3 = new Student();
      s3.st_Id = 103;  s3.name = "Arjun";  s3.age = 21;
      s1.displayDetails();  s2.displayDetails();   s3.displayDetails();    }}
```

Output:

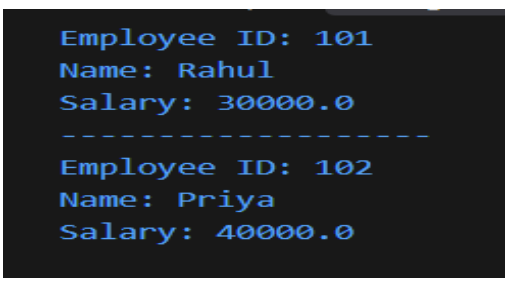
```
Student ID: 101
Name: Rahul
Age: 20
--------------------
Student ID: 102
Name: Priya
Age: 19
--------------------
Student ID: 103
Name: Arjun
Age: 21
```

2. Create an Employee class with employee ID, name and salary. Initialize the values using a constructor and display employee details.

Code:

```
class Employee {
    int employeeId;
    String name;
    double salary;
    Employee(int id, String n, double s) {
        employeeId = id;  name = n;  salary = s;
    }
    void displayDetails() {
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Name: " + name);
        System.out.println("Salary: " + salary);
    }
    public static void main(String[] args) {
        Employee e1 = new Employee(101, "Rahul", 30000);
        Employee e2 = new Employee(102, "Priya", 40000);
        e1.displayDetails();
        System.out.println("--------------------");
        e2.displayDetails();
    }
}
```

Output:



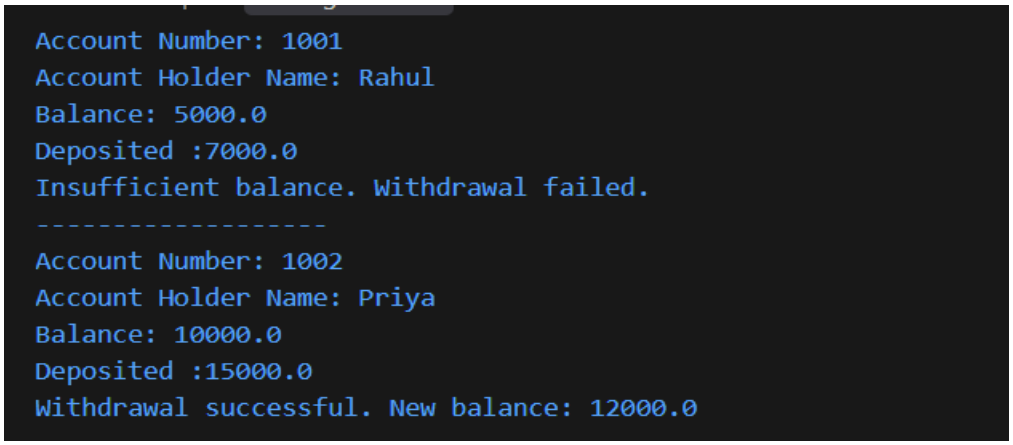
3. Create a Bank Account class with account number and balance. Implement deposit and withdrawal methods.

Code:

```
public class BankAccount {
    int accountNumber;
    String accountHolderName;
    double balance;
    BankAccount(int accNum, String accHolderName, double bal) {
        accountNumber = accNum;
        accountHolderName = accHolderName;
        balance = bal;    }
    void displayDetails() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Account Holder Name: " + accountHolderName);
        System.out.println("Balance: " + balance);    }
    void deposit(double amount) {
        balance += amount;
        System.out.println("Deposited :" + balance);    }
    void withdrawal(double amount) {
        if (amount <= balance) {
            balance -= amount;
            System.out.println("Withdrawal successful. New balance: " + balance);
        } else {
            System.out.println("Insufficient balance. Withdrawal failed.");        }}
    public static void main(String[] args) {
        BankAccount b1 = new BankAccount(1001, "Rahul", 5000.0);
        BankAccount b2 = new BankAccount(1002, "Priya", 10000.0);
        b1.displayDetails();
        b1.deposit(2000.0);
```

```
b1.withdrawal(12000.0);
System.out.println("--------------------");
b2.displayDetails();
b2.deposit(5000.0);
b2.withdrawal(3000.0);     }}
```

Output:



4. Create a Rectangle class with length and width. Calculate and display its area and perimeter.

Code:

```
class Rectangle {
double length;
double width;
Rectangle(double l, double w) {
length = l;
width = w;    }
void calculateArea() {
double area = length * width;
System.out.println("Area: " + area);    }
void calculatePerimeter() {
 double perimeter = 2 * (length + width);
  System.out.println("Perimeter: " + perimeter);    }
```

```
public static void main(String[] args) {
Rectangle r1 = new Rectangle(10, 5);
r1.calculateArea();
r1.calculatePerimeter();    }     }
```

Output:

```
Area: 50.0
Perimeter: 30.0
```

5. Create a Book class with ISBN, title and price. Initialize the values using a parameterized constructor.

Code:

```
class Book {
  String ISBN;    String title;    double price;
  Book(String i, String t, double p) {
    ISBN = i;      title = t;      price = p;    }
  void displayDetails() {
    System.out.println("ISBN: " + ISBN);
    System.out.println("Title: " + title);
    System.out.println("Price: " + price);    }
  public static void main(String[] args) {
    Book b1 = new Book("9781234567890", "Java Programming", 550);
    b1.displayDetails();    }    }
```

Output:

```
ISBN: 9781234567890
Title: Java Programming
Price: 550.0
```